

"The whole reason we are investing into the cell line is to be able to create the cell that we want in India"

–PANKAJ SHARMA
Co-founder, Log9

range from $6 billion (Rs 49,890 crore), on the conservative side, to $15 billion (Rs 1.25 lakh crore), if things move at a faster clip. The global market by then is expected to exceed $150 billion (Rs 12.47 lakh crore) annually. "India...could represent up to 13 per cent of global battery demand by 2030," the think-tank said in a February 2022 report.

But this demand for batteries will have to be met through domestic manufacturing to ensure the country's energy security. In 2022, the government announced a Rs 18,100 crore production-linked incentive scheme for the manufacture of advanced chemistry cells. India, currently the third largest oil importing country in the world, hopes to save Rs 2 lakh crore on its oil bill over the scheme duration of five years.

WHAT INDIA NEEDS TO MASTER

The basic unit of a battery is a cell. The Li-ion batteries used in electric cars comprise advanced chemistry cells, which are stacked together with sophisticated heat management systems running on electronic chips and software. Currently, nearly the entire domestic demand for Li-ion cells is being met through imports.

Globally, China accounts for 78 per cent of Li-ion battery manufacturing capacity, followed by the US and Europe at 8 per cent and 7 per cent, respectively. Getting a foothold in this supply chain is critical, and strewn with challenges. First, minerals such as lithium and cobalt that serve as raw material for these batteries aren't mined in India.

THE CHANGE-MAKERS

LOG9 MATERIALS

- This April, the Bengaluru-based start-up switched on the only Li-ion cell line by a battery-maker in India
- In the cell line, raw materials such as titanate and graphite are coated on foils, jelly-rolled into a cylindrical cell and the electrolyte filled in
- Log9, incubated at IIT Roorkee in 2015, develops standard and larger format cells which go into batteries for three- and two-wheelers
- With focus on fast-charging, Log9 is betting on a battery chemistry called lithium-titanate
- Next up: aluminium fuel cells in which a block of aluminium will serve as the fuel, reacting with oxygen to produce energy and leaving behind aluminium hydroxide from which aluminium can be smelted

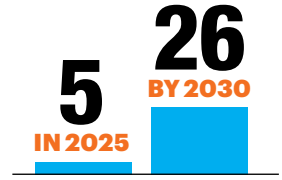
CYGNI ENERGY

- Incubated at IIT Madras in 2014, Cygni first ventured into solar energy to build direct current convertors before pivoting to Li-ion batteries and renewable energy storage
- It works with 3 of the top 10 electric vehicle manufacturers in India
- Battery packs are a semiconductor- and software-intensive effort. There are over 80 components that go inside a battery pack, says founder Venkat Rajaraman
- Cygni's focus is on building the intellectual property for battery management systems
- Last year, it raised $12.5 million (Rs 103.9 crore) for a 1.2 GWh gigafactory in Hyderabad that will commence operations in early 2024

50 GWh
The manufacturing capacity the current PLI scheme aims to create, by attracting direct investment of Rs 45,000 crore

GIGAFACTORIES INDIA WILL NEED

To meet the demand for batteries. Each factory with a manufacturing capacity of 10 GWh per year



But there's an opportunity to recycle dead batteries for these critical elements. Urban mining, it is estimated, can recover enough lithium to produce 80-gigawatt hours of batteries by 2030. In 2022, the government brought in new battery waste management rules. But building an efficient waste recycling industry is key.

Battery technology involves advanced chemistries and electronics, and India has to rapidly create an ecosystem for that, if it has to do justice to its quest for higher energy density. The key approaches in this direction are to replace today's graphite anodes with lithium metal and liquid electrolytes with solids. Recharging lithium metal has been a notoriously old problem but an exciting field of research today. When cracked, it can open the door for new chemistries such as lithium sulphur and help push India's big green future plan. ■